Healthcare Insurance Analysis

Business Scenario

A significant public health concern is the rising cost of healthcare. Therefore, it's crucial to be able to predict future costs and gain a solid understanding of their causes. The insurance industry must also take this analysis seriously. This analysis may be used by healthcare insurance providers to make a variety of strategic and tactical decisions.

Problem statement

The objective of this project is to predict patients' healthcare costs and to identify factors contributing to this prediction. It will also be useful to learn the interdependencies of different factors and comprehend the significance of various tools at various stages of the healthcare cost prediction process.

Dataset Description

Three datasets were used in this project.

Hospitalisation Details Dataset

This dataset contains patient-level hospitalization information. Important fields include:

- Customer ID
- Year, month, and date of birth
- Number of children
- Hospitalization charges
- Hospital tier
- City tier
- State ID

Medical Examinations Dataset

This dataset contains medical and health-related information. Important fields include:

- Customer ID
- BMI

- HBA1C
- Heart Issues
- Any Transplants
- Cancer history
- Number of Major Surgeries
- Smoker

Names Dataset

This dataset contains:

- Customer ID
- Beneficiary name

Data Importing and Merging

Three sets of data were used for this project. These datasets are hospitalization, medical examinations, and names. They were uploaded with the help of Python packages such as pandas, numpy, matplotlib, and seaborn.

These three datasets were combined using Customer ID, which was a common field for all the sets. Firstly, the hospitalization dataset was combined with the medical examination set. Secondly, the names set was combined with the other two using the same Customer ID.

After the combination process, the resulting dataset consisted of 2,335 records and 17 fields. It gave a full picture of each patient since the hospital, medical, and name data were all included in one dataset.

It was significant since the hospitalization dataset included only cost, birth, hospital, city, and state data. The medical examination data included factors that were related to patients' health, their BMI, HBA1C, heart problems, cancer background, surgery number, and whether the person smoked.

Missing Value Checking

After merging the data, the presence of missing values was checked with info() and isnull().sum(). The results showed that there were no any direct missing values in the dataset. All columns had full non-null values.

It is a good thing, as the absence of missing values means that there was no need to impute any missing values. However, lack of null values does not mean that the data is totally clean, as some columns had trivial values such as ?.

Trivial Values Checking and Their Removal

The dataset was checked for the existence of trivial values, especially question marks (?). Such trivial values cannot provide any information. The percentage of the rows which have trivial values was calculated as 0.43%. As this percentage is very small, the rows were removed from the dataset. The total amount of rows became 2,325, while the total amount of columns remained the same (17).

Categorical Variables Transformation

Categorical variables were transformed in order to use them in further analysis and machine learning. Nominal categorical variables, such as Heart Issues, Any Transplants, Cancer history and smoker, were transformed with label encoding method. These categorical variables contained textual information (Yes and No), and after transformation they become numerical. Ordinal categorical variables, such as Hospital tier and City tier were cleaned and transformed to numbers. For instance, values like tier - 1, tier - 2, tier - 3 became 1, 2, and 3 correspondingly. It was necessary because the hospital tier and the city tier are ordinal variables, and such terms as tier - 1, tier - 2, tier - 3 are not just labels, but ordered categories.

State ID Dummy Variables

There were many states in the State ID column. However, as mentioned in the task, only R1011, R1012 and R1013 states should be further investigated. The creation of dummy variables for all states would generate unnecessary amount of predictors.

That is why only three dummy variables were created,

- State_R1011
- State_R1012
- State_R1013

Every dummy variable gets 1 if the patient belongs to the corresponding state, and 0 otherwise.

Such approach was appropriate, as it helped to reduce the number of predictors while keeping the most important state information for modelling.

Cleaning Number of Major Surgeries

The column "NumberOfMajorSurgeries" contains some strings. However, since this column is a numerical column, it is necessary to take only numeric part of the strings. After this, this column was converted into integer type. Missing values and non-numeric values became 0s.

The clean surgery count was divided as follows:

• 0 major surgeries: 1,070 patients

• 1 major surgery: 961 patients

• 2 major surgeries: 272 patients

• 3 major surgeries: 22 patients

As one can see, most patients had either 0 major surgeries, or 1 major surgery.

Age Calculation

Age is considered as one of the key factors of healthcare cost prediction. There were separate columns which provided the year, month and date of birth. Since there were text values in the month column (Jan, Feb, Jul, Nov), the month values were firstly transformed into numeric month values. After this a new column DOB was created with the help of year, month and date columns. Then age was calculated for every patient with fixed analysis date. Age calculation resulted in realistic age values. For example, those who were born in 1992, their ages became about 33 years old, while those who were born in 1998 became 27 years old. This transformation was necessary, as age can significantly influence the cost of hospitalization and medical risks.

Gender Extraction

Gender variable was generated from the beneficiary name column. The salutation in every name was used to identify gender. The following rule was applied: Mr. and Master became Male, Ms., Mrs., and Miss became Female. So, a new Gender variable was created. This variable was encoded into numbers to use it in further visual and machine learning analysis. This transformation was helpful, as the initial dataset did not contain Gender variable, but the name column contained necessary information.

Final Cleaned Dataset

After the cleaning and feature generation, the dataset contained 2,325 rows and 24 columns.

The new columns include:

- State_R1011
- State_R1012
- State_R1013
- month_num
- DOB
- Age
- Gender

This dataset is ready to perform exploratory data analysis, hypothesis testing and machine learning.

Exploratory Data Analysis

Exploratory data analysis was conducted to understand cost distribution of hospitalizations and the difference in cost across different categories of patients and hospitals.

Cost Distribution

The histogram shows that the cost distribution is right-skewed.

The great majority of patients have low to mid-range hospitalization costs. However, several patients have a very high cost. It means that the dataset has high-cost outliers. It is crucial for insurance analysis as a relatively small number of expensive patients can significantly affect the overall claim cost.

Boxplot for Hospitalization Cost

Boxplot demonstrates that many patients have low to mid-range cost but there are several high-cost outliers. They mean that some patients have much higher cost of hospitalization compared to the rest of the sample. It is significant for insurance providers as those patients might be considered risky ones.

Swarmplot by Gender

Swarmplot was created to find out the cost distribution of hospitalizations for males and females. It was shown that both males and females have wide range of cost of hospitalization. There are high-cost cases among patients from both groups. It means that gender is not the main cost driver, however, it provides additional information when combined with other variable.

Cost Distribution by Hospital Tier and Gender

Boxplot was used to find out the cost distribution by hospital tier and gender. It was revealed that the hospital tier is influential factor for hospitalization cost. Tier-1 hospitals have the highest cost, tier-2 and tier-3 hospitals have lower cost. The mentioned pattern was observed for patients of both genders. The summary table shows that for males tier-1 hospitals have approximately 30,004 cost while tier-2 hospitals have approximately 12,365 cost and tier-3 hospitals have approximately 8,637 cost. For females tier-1 hospitals have approximately 30,416 cost while tier-2 hospitals have approximately 11,526 cost and tier-3 hospitals have approximately 10,636 cost. Therefore, the hospital tier is stronger cost driver than gender.

Radar Chart for Median Cost by Hospital Tier

Radar chart was created to find the median cost of hospitalization for each hospital tier.

The median cost is:

Tier 1: approximately 32,097

Tier 2: approximately 7,169

Tier 3: approximately 10,677

It is clearly shown in the radar chart that tier-1 hospitals have the highest median cost of hospitalization. Therefore, the findings that tier-1 hospitals have higher cost are supported.

Frequency Table and Stacked Bar Chart

Frequency table and stacked bar chart were created to find out the number of patients across the city tiers and hospital tiers.

The frequency table shows:

City tier 1: 85 patients in tier-1 hospitals, 403 in tier-2 hospitals, and 241 in tier-3 hospitals

City tier 2: 106 patients in tier-1 hospitals, 479 in tier-2 hospitals, and 222 in tier-3 hospitals

City tier 3: 109 patients in tier-1 hospitals, 452 in tier-2 hospitals, and 228 in tier-3 hospitals

It means that tier-2 hospitals have the greatest number of patients' records while tier-1 hospitals have fewer records but their cost is much higher.

Hypothesis Tests

The four hypothesis tests were performed in order to understand whether some groups have statistically different hospitalization costs.

1. Hospital Tier and Hospitalization Cost

The first null hypothesis stated that the average hospitalization cost for three hospital tiers is not significantly different. ANOVA test was performed. The result was:

F-statistic: 493.99

P-value: 1.77e-179

As p-value was less than 0.05 the null hypothesis was rejected.

It means that average hospitalization cost is significantly different across hospital tiers. The hospital tier is the statistically significant factor for hospitalization cost. Therefore, it confirms the earlier finding that tier-1 hospitals have much higher cost.

2. City Tier and Hospitalization Cost

The second null hypothesis stated that the average hospitalization cost for three city tiers is not significantly different. ANOVA test was performed. The result was:

F-statistic: 1.45

P-value: 0.234

As p-value was greater than 0.05 the null hypothesis was not rejected. It means that the average hospitalization cost for the three city tiers is not significantly different. The city tier is not the strong statistical factor of hospitalization cost in this dataset.

3. Smoking Status and Hospitalization Cost

The third null hypothesis stated that the average hospitalization cost for smokers is not significantly different from the average hospitalization cost for non-smokers.

The result was:

T-statistic: 56.32

P-value: 6.70e-238

As p-value was less than 0.05 the null hypothesis was rejected. It means that the cost of hospitalization for smokers and non-smokers is significantly different. Therefore, smoking status is important factor in predicting healthcare costs.

4. Smoking and Heart Issues

The fourth null hypothesis stated that smoking and heart issues are independent. chi-square test was performed. The result was:

Chi-square value: 0.086

P-value: 0.769

Degrees of freedom: 1

As p-value was greater than 0.05 the null hypothesis was not rejected. It means that smoking and heart issues are independent in this dataset. In other words, this dataset does not show any statistically significant relationship between smoking status and heart issues.

Machine Learning Results and Interpretation

The machine learning task was used to predict hospitalization cost. The target variable was charges.

Correlation Heatmap

Based on the heatmap, it is possible to see that the variable 'smoker' had the greatest positive correlation with hospitalization cost. At the same time, such predictors as BMI and age also demonstrated positive relations with the cost. On the other hand, hospital tier had a negative relation since it was coded with numbers 1, 2, and 3 where the cost in the first tier was the highest but its numeric code was the lowest. It means that as the hospital tier changed from 1 to 3, the cost tended to decrease. It helped to identify the most relevant predictors before the model building.

Model Features

The following predictors were chosen for modeling:

Age

Children

BMI

HBA1C

Number of Major Surgeries

Hospital tier

City tier

Smoker

Heart Issues

Any Transplants

Cancer history

State_R1011

State_R1012

State_R1013

Gender

The target variable was hospitalization cost (charges).

Stratified 5-Fold Cross-Validation

A stratified 5-fold cross-validation technique was used. As the problem was about regression, the target variable (charges) was first split into cost bins using quantiles. Based on them, the folds were created.

The distribution of the folds was equal:

Fold 1: 465 observations

Fold 2: 465 observations

Fold 3: 465 observations

Fold 4: 465 observations

Fold 5: 465 observations

Such an approach helped to balance the folds.

Pipeline and Standardization

A machine learning pipeline was applied in order to make the workflow more systematic. The pipeline included such stages as preprocessing and modeling.

Firstly, numerical features were normalized using the StandardScaler technique. At the same time, gender was considered to be a categorical feature and was transformed using OneHotEncoder. The pipeline application helped to prevent data leakage.

Ridge Regression Model

As the first model, ridge regression was used as a regularized linear regression model. Hyperparameter tuning was done using different alpha parameters.

The best ridge regression parameter was:

alpha = 0.01

Ridge Regression achieved a mean absolute error (MAE) of 2,741.30.

It means that the average error of Ridge model prediction is 2,741 cost units.

Gradient Boosting Regression Model

In addition, gradient boosting regressor was modeled and tuned. Hyperparameter tuning was done using different values for number of estimators, learning rate, and maximum depth. The MAE of the Gradient Boosting model is 2,078.38. It is less than the ridge regression error. In this way, gradient boosting showed better results and became the best model.

Model Comparison

Model comparison produced the following results:

Ridge Regression MAE: 2,741.30

Gradient Boosting MAE: 2,078.38

Since Gradient Boosting had the lowest MAE, it was chosen as the final model. Therefore, it can be concluded that a nonlinear model turned out to be more appropriate for this dataset rather than linear one. Cost prediction in the healthcare sector may require the consideration of many non-linear relations between the variables. It explains why gradient boosting is capable of capturing more patterns.

Feature Importance

The feature importance produced by gradient boosting model is shown below:

Smoker: 0.751

BMI: 0.112

Age: 0.093

Hospital tier: 0.021

State_R1011: 0.007

The result obtained shows that smoking status was the most influential predictor in the model. In addition, BMI and age were also influential medical risk factors. In addition, the hospital tier influenced cost prediction.

Some of the variables had much smaller feature importance such as heart issues, city tier, any transplants, and cancer history. They were not considered important enough for prediction.

Redundant Variables

Variables whose feature importance is less than 0.01 can be considered redundant or weak predictors. They include:

State_R1011

Children

State_R1013

HBA1C

Number of Major Surgeries

Gender

Cancer history

State_R1012

Any Transplants

City tier

Heart Issues

They are less influential than smoker, BMI, age, and hospital tier in this model.

Case Scenario Prediction

The case scenario for the patient named Ms. Jayna Christopher was created. Her profile includes the following information:

Date of Birth: 28 December 1988

Height: 170 cm

Weight: 85 kg

Approximate BMI: 29.41

Two children

Tier-1 city

State ID: R1011

HBA1C: 5.8

Smoker

No heart issues

No transplants

No major surgeries

Cancer history

Tier-1 hospital

Female

The best model (gradient boosting) was used for the hospitalization cost prediction for this patient profile.

Part 2 SQL

In order to analyze the healthcare insurance using SQL, some structured operations were performed on the database. The main aim of the SQL part was to join the tables, clean the required column, add primary keys, and obtain useful information regarding hospitalization costs.

Task 1: Merging the Tables

The first SQL task was to merge the hospitalization and medical examination tables. The common column identified for merging was Customer ID, as this column uniquely represents each patient in both tables.

Before merging, Customer ID was checked for null, blank, invalid, and duplicate values. This step was necessary because a primary key must be unique and cannot contain missing values. After checking and cleaning the Customer ID column, the datatype was changed to VARCHAR(50) so that it could be used as a primary key. A primary key constraint was then added to both the hospitalisation_details and medical_examinations tables.

After this, both tables were merged using an inner join on Customer ID. This produced a combined dataset containing hospitalization details such as age-related birth information, children, charges, hospital tier, city tier, and state ID, along with medical details such as BMI, HBA1C, heart issues, transplant status, cancer history, number of major surgeries, and smoking status.

This merged result helped create a complete view of each patient and allowed further analysis of the factors influencing hospitalization cost.

Task 2: Diabetic Patients with Heart Problems

The second SQL query retrieved information about people who were diabetic and also had heart problems. In this query, patients were considered diabetic when their HBA1C value was greater than 6.5, and heart problems were identified where the Heart Issues value was Yes.

The query calculated the total number of such patients, their average age, average number of dependent children, average BMI, and average hospitalization cost.

The output showed that there were **323 diabetic patients with heart problems**. Their average age was approximately **52.41 years**, the average number of children was **1.02**, the average BMI was **31.36**, and the average hospitalization cost was approximately **16,340.40**.

This result suggests that diabetic patients with heart problems represent an important risk group for insurance providers because they have notable healthcare costs and may require closer medical and financial assessment.

Task 3: Average Hospitalization Cost by Hospital Tier and City Tier

The third SQL query calculated the average hospitalization cost for each combination of hospital tier and city tier. This helped compare how costs differed across different hospital quality levels and city development levels.

The result showed that tier-1 hospitals had the highest average hospitalization costs across all city tiers. For example, tier-1 hospitals had average costs of approximately 29,160.76 in city tier 1, 29,014.50 in city tier 2, and 31,976.12 in city tier 3.

Tier-2 hospitals had lower average costs, ranging from approximately 11,515.41 to 12,093.67. Tier-3 hospitals had the lowest average costs, ranging from approximately 9,283.43 to 9,812.84.

This result shows that hospital tier is a stronger cost driver than city tier. Tier-1 hospitals are associated with much higher hospitalization costs compared with tier-2 and tier-3 hospitals.

Task 4: Major Surgery with Cancer History

The fourth SQL query counted the number of people who had undergone major surgery and also had a history of cancer. The query selected patients where the number of major surgeries was greater than zero and cancer history was marked as Yes.

The result showed that **391 people** had both a history of cancer and at least one major surgery.

This is useful for insurance analysis because patients with cancer history and previous major surgeries may be associated with higher medical risk. Such patients may need more detailed cost prediction and risk assessment.

Task 5: Tier-1 Hospital Records by State

The fifth SQL query counted the number of tier-1 hospital records in each state. Since the dataset does not contain a separate hospital ID column, this result represents the number of patient records associated with tier-1 hospitals in each state.

The result showed that **R1011** had the highest number of tier-1 hospital records, with **116 records**. This was followed by **R1013** with **68 records** and **R1012** with **62 records**. Other states had much smaller counts.

This result indicates that tier-1 hospital usage is not equally distributed across states. R1011 appears to be the most important state for tier-1 hospital analysis in this dataset.

Tableau Dashboard Results

A Tableau dashboard was created to present the main findings from the healthcare insurance analysis in a visual and business-friendly format. The dashboard focused on data storytelling and included KPI cards, bar charts, a histogram, a stacked bar chart, and a hospital tier filter.

The dashboard title was Healthcare Insurance Cost Analysis Dashboard. A short story text was added to explain that the dashboard analyses hospitalization costs using hospital, city, state, and medical risk factors. The aim of the dashboard was to help healthcare insurance providers understand the main drivers of patient hospitalization costs.

KPI 1: Total Patients

The first KPI card showed the total number of patients in the cleaned dataset. The value was 2,325 patients.

This confirms the number of records used for the dashboard after data cleaning and removal of trivial values. It gives the viewer a quick understanding of the dataset size.

KPI 2: Average Hospitalization Cost

The second KPI card showed the overall average hospitalization cost. The average hospitalization cost was approximately 13,522.

This value gives a high-level summary of the typical hospitalization cost across the full dataset. It acts as a reference point when comparing costs across hospital tiers, city tiers, states, and smoking status.

Average Cost by Hospital Tier

The dashboard showed that hospital tier had a major impact on hospitalization cost. Tier-1 hospitals had the highest average cost at approximately 30,132. Tier-2 hospitals had an average cost of approximately 11,876, while tier-3 hospitals had the lowest average cost at approximately 9,487.

This result clearly shows that tier-1 hospitals are much more expensive than tier-2 and tier-3 hospitals. This may be because tier-1 hospitals are higher-quality hospitals, may provide more advanced treatment, or may be located in areas with higher healthcare expenses.

For insurance providers, hospital tier is therefore an important factor when predicting hospitalization cost and assessing policy risk.

Average Cost by Smoking Status

The dashboard also showed a strong difference in average hospitalization cost between smokers and non-smokers. Smokers had an average hospitalization cost of approximately 32,867, while non-smokers had an average cost of approximately 8,409.

This is one of the strongest findings in the dashboard. It suggests that smoking status is a major medical risk factor and is strongly associated with higher hospitalization costs.

For insurance companies, smoking status should be considered an important variable in healthcare cost prediction and risk-based pricing.

Average Cost by City Tier

The dashboard compared average hospitalization costs across city tiers. City tier 3 had the highest average cost at approximately 14,045, followed by city tier 2 at approximately 13,472, and city tier 1 at approximately 13,010.

Although there are some differences between city tiers, the differences are smaller compared with the differences seen across hospital tiers and smoking status. This suggests that city tier may have some influence on cost, but it is not as strong as hospital tier or smoking status.

Average Cost by State

The state-level chart showed average hospitalization costs across different states. The highest average cost was seen in R1011, with an average cost of approximately 19,466. Other states had lower average costs, with several states ranging between around 8,789 and 14,806.

This chart shows that hospitalization cost varies by state. Some states may have higher healthcare costs because of differences in hospital usage, patient characteristics, or medical risk

factors. R1011 appears to be an important state for further investigation because it has the highest average hospitalization cost.

Distribution of Hospitalization Cost

The histogram showed the distribution of hospitalization costs. Most patients had hospitalization costs below 15K. The most frequent cost ranges were 0K to 5K, 5K to 10K, and 10K to 15K. Only a small number of patients had very high costs above 40K.

This shows that the hospitalization cost distribution is right-skewed. Most patients have relatively low or moderate costs, but a small number of high-cost patients increase the overall average cost.

This is important for insurance providers because high-cost outliers can strongly affect total claims and financial risk.

Patient Count by City and Hospital Tier

The stacked bar chart showed the distribution of patients across city tiers and hospital tiers. The chart showed that tier-2 hospitals had the highest number of patient records across city tiers. For example, tier-2 hospital counts were 403, 479, and 452 across the different city tiers.

Tier-3 hospitals also had a large number of records, with counts such as 241, 222, and 228. Tier-1 hospitals had fewer patient records compared with the other hospital tiers.

This result shows that most patients are linked with tier-2 and tier-3 hospitals, while tier-1 hospitals have fewer records but much higher average costs.

Hospital Tier Filter

A hospital tier filter was added to the dashboard. This allows users to filter the dashboard by hospital tier and view how costs and patient distributions change for tier-1, tier-2, and tier-3 hospitals.

The filter improves interactivity and allows business users to explore cost patterns more easily.

Summary of Tableau Findings

The Tableau dashboard highlights several important findings:

- The cleaned dataset contains 2,325 patients.
- The average hospitalization cost is approximately 13,522.
- Tier-1 hospitals have the highest average hospitalization cost.
- Smokers have much higher average hospitalization costs than non-smokers.

- City tier differences are present but smaller than hospital tier and smoking differences.
- R1011 has the highest average cost among states.
- Most patients have costs below 15K, but a few high-cost cases create outliers.
- Most patient records are associated with tier-2 and tier-3 hospitals.

Overall, the Tableau dashboard supports the conclusion that hospital tier and smoking status are the most important cost drivers. The dashboard also helps insurance providers identify high-cost patient groups and make better pricing, risk management, and healthcare planning decisions.